

Worksheet 1B: Fractions in the Wild

Annotated Revision Guide

AIML Foundations Mathematics

Dublin and D'un Laoghaire ETB

Instructor: Josh Aaron

Part A: Physics — Special Relativity (Problems 1–4)

Concept. The **Lorentz factor** $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$ appears throughout special

relativity. Its behaviour is entirely determined by the fraction $\frac{v^2}{c^2}$: as $v \rightarrow c$, this ratio approaches 1, the square root approaches 0, and γ blows up. All relativistic quantities (energy, momentum, time) involve this same fraction pattern.

Method. For each problem: substitute the given value of v/c , evaluate the fraction under the radical, then analyse what happens as v approaches c . No calculation of c itself is needed — only the *ratio* v/c matters.

Watch out. $v = c$ is not just hard to reach — it is *impossible* for any massive object. The formulas show why: energy and momentum would both require $\gamma \rightarrow \infty$, meaning infinite energy to accelerate any mass to c .

Where this leads. The factor $\sqrt{1 - v^2/c^2}$ reappears in the metric tensor of special relativity and in the Lorentz transformation. In ML, analogous "normalisation denominators" appear in attention mechanisms (Worksheet 1B problem 28) and batch normalisation.

Explore further. Mermin, *Space and Time in Special Relativity* (1968) is the most readable introduction. The entire structure of special relativity follows from two postulates; the Lorentz factor is a consequence, not an assumption.

1. Lorentz factor $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$.

(a) $v = 0$. $\frac{v^2}{c^2} = 0$; $\gamma = \frac{1}{\sqrt{1}} = 1$. (Object at rest: no time dilation.)

(b) $v = 0.5c$. $\frac{v^2}{c^2} = 0.25$; $1 - 0.25 = 0.75$; $\gamma = \frac{1}{\sqrt{0.75}} \approx 1.155$.

(c) As $v \rightarrow c$. $\frac{v^2}{c^2} \rightarrow 1$.

(d) As $v \rightarrow c$. $1 - \frac{v^2}{c^2} \rightarrow 0$.

(e) As $v \rightarrow c$. $\sqrt{\cdot} \rightarrow 0$, so $\gamma = \frac{1}{\sqrt{\cdot}} \rightarrow \infty$.

2. Time dilation: $t_{\text{moving}} = t_{\text{stationary}} \times \sqrt{1 - v^2/c^2}$.

(a) $v = 0$. $\sqrt{1 - 0} = 1$; $t_{\text{moving}} = t_{\text{stationary}}$. Clocks agree.

(b) As $v \rightarrow c$. $\sqrt{1 - v^2/c^2} \rightarrow 0$, so $t_{\text{moving}} \rightarrow 0$.

(c) Moving clocks run **slower**: a moving clock reads less elapsed time than a stationary one.

3. Relativistic energy $E = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$.

- (a) $v = 0$. Denominator = 1; $E = mc^2$. This is the famous rest energy.
- (b) As $v \rightarrow c$. Denominator $\rightarrow 0$; $E \rightarrow \infty$.
- (c) Accelerating a massive object to c would require infinite energy. Therefore nothing with mass can reach light speed.

4. Relativistic momentum $p = \frac{mv}{\sqrt{1 - v^2/c^2}}$.

- (a) **Small** v . Denominator ≈ 1 ; $p \approx mv$ — classical momentum.
- (b) As $v \rightarrow c$. Denominator $\rightarrow 0$; $p \rightarrow \infty$.

Look ahead: Newton's $F = ma$ is the low-velocity limit of relativistic mechanics. All of classical physics emerges when $v \ll c$.

Part B: Probability — Bayes' Theorem (Problems 5–7)

Concept. Bayes' theorem: $P(H|E) = \frac{P(E|H) P(H)}{P(E)}$. It says: to find the probability of a hypothesis given evidence, weight the prior probability by the likelihood, then normalise by how probable the evidence is overall. The key insight is that the denominator $P(E)$ includes *all* ways the evidence could occur, including false positives.

Method. Analyse the fraction qualitatively: small numerator (rare disease, or weak test) $\div$ large denominator (common evidence) = small posterior probability. No calculation is needed for most of these problems — just reason about which parts of the fraction are large or small.

Watch out. Base rate neglect: humans naturally focus on the test accuracy and ignore $P(H)$. A 99% accurate test for a 1-in-10,000 condition still gives mostly false positives. The fraction makes this precise: the numerator is tiny even with high $P(E|H)$.

Where this leads. Bayes' theorem is the foundation of **Bayesian machine learning**: updating model beliefs given data. Naive Bayes classifiers, Bayesian neural networks, and MCMC sampling all apply this same fraction update.

Explore further. Silver, *The Signal and the Noise* (2012), Ch. 8, gives excellent intuitive examples. For rigour: Jaynes, *Probability Theory: The Logic of Science* (2003).

5. Medical test: $P(\text{disease}) = 0.01$, sensitivity = 0.99, false positive rate = 0.05.

- (a) **Small numerator.** Small $P(\text{disease}) \Rightarrow$ small numerator $P(E|H) \cdot P(H) = 0.99 \times 0.01 = 0.0099$.

(b) **Large denominator.** High false positive rate raises $P(\text{positive})$: $P(E) \approx 0.01 \times 0.99 + 0.99 \times 0.05 \approx 0.0099 + 0.0495 = 0.0594$.

(c) $P(\text{disease}|\text{positive}) = \frac{0.0099}{0.0594} \approx 17\%$. Despite 99% sensitivity, a positive test means only a 17% chance of disease!

Note: This counterintuitive result is called the false positive paradox. It is why mass screening for rare diseases requires very careful interpretation.

6. Common condition: $P(\text{condition}) = 0.5$.

(a) Numerator $\propto P(H) = 0.5$ — much larger than the rare-disease case.

(b) A positive test is far more meaningful for a common condition: the prior is not fighting against you.

7. What happens if (a) $P(E)$ is very small; (b) $P(E) \approx 1$?

(a) Dividing by a small $P(E)$ gives a large posterior — rare evidence is very informative.

(b) Dividing by $P(E) \approx 1$ leaves the posterior close to the likelihood ratio — common evidence provides little update.

Part C: Information Theory — Entropy (Problems 8–10)

Concept. Surprise: $I(p) = \log_2\left(\frac{1}{p}\right) = -\log_2 p$ bits. Rare events ($p \approx 0$) are highly surprising; certain events ($p = 1$) carry zero information. **Shannon entropy** $H = -\sum p_i \log_2 p_i$ is the average surprise across all outcomes. High entropy means unpredictability; low entropy means predictability.

Method. For problems 8–9: reason about the fraction $\frac{1}{p}$. As $p \rightarrow 0$, $\frac{1}{p} \rightarrow \infty$, so $\log_2 \frac{1}{p} \rightarrow \infty$. As $p \rightarrow 1$, $\frac{1}{p} \rightarrow 1$, so $\log_2(1) = 0$.

Watch out. Entropy is about *unpredictability*, not disorder. A maximally compressed file has high entropy — every bit is "surprising". A file of all zeros has zero entropy — perfectly predictable.

Where this leads. The **cross-entropy loss** used to train neural network classifiers is $-\sum y_i \log \hat{p}_i$, a direct application of surprise. Minimising cross-entropy is equivalent to maximising likelihood.

Explore further. Shannon, *A Mathematical Theory of Communication* (1948) — the paper that founded information theory. Available free online. Read the first 10 pages.

8. Surprise $= \log_2\left(\frac{1}{P(\text{event})}\right)$.

(a,b) **Likely event.** $P \approx 1 \Rightarrow \frac{1}{P} \approx 1 \Rightarrow \log_2(1) = 0$. Low surprise.

(c,d) **Unlikely event.** $P \approx 0 \Rightarrow \frac{1}{P} \rightarrow \infty \Rightarrow \log_2 \rightarrow \infty$. High surprise.

(e) Rare events are **more** surprising.

9. Fair coin vs biased coin ($P(\text{heads}) = 0.99$).

(a) Biased coin is more predictable.

(b) Biased coin has lower entropy: nearly all outcomes are "heads", little average surprise.

(c) Biased coin can be compressed more efficiently: a long string of results is almost all "heads", so a short code suffices.

10. Compression ratio = $\frac{\text{compressed size}}{\text{original size}}$.

(a) Ratio close to 0 means good compression (much smaller).

(b) Ratio = 1: no compression at all.

(c) Yes — if the compressed file is *larger* than the original. This happens when data is already maximally random (high entropy).

Part D: Machine Learning — Sigmoid, Softmax, Gradient Descent (Problems 11–16)

Concept. These fractions are the building blocks of neural networks. The **sigmoid** $\sigma(x) = \frac{1}{1+e^{-x}}$ maps any real number to $(0, 1)$. The **softmax** generalises this to multi-class probabilities. The **gradient descent** update uses a fraction (the gradient) to adjust model parameters.

Method. For sigmoid: substitute the given x , evaluate e^{-x} , compute $1 + e^{-x}$, take the reciprocal. For softmax: the denominator normalises all outputs to sum to 1. For batch normalisation: understand the formula as a z -score.

Watch out. The sigmoid never reaches 0 or 1 for finite x . It is asymptotic. Similarly, softmax values never reach exactly 0 or 1. This matters for numerical stability in training.

Where this leads. The sigmoid is the "squashing" function for binary classification. Softmax is for multi-class. Both are differentiable, which is why gradient descent works. The derivative of sigmoid is $\sigma(x)(1 - \sigma(x))$ — another elegant fraction.

Explore further. Goodfellow, Bengio & Courville, *Deep Learning* (2016), Chapters 6–7. Free at deeplearningbook.org.

11. Sigmoid: $\sigma(x) = \frac{1}{1 + e^{-x}}$.

(a) $x = 0$. $e^0 = 1$; $\sigma(0) = \frac{1}{1+1} = 0.5$.

(b) $x \rightarrow +\infty$. $e^{-x} \rightarrow 0$; $\sigma \rightarrow \frac{1}{1+0} = 1$.

(c) $x \rightarrow -\infty$. $e^{-x} \rightarrow \infty$; $\sigma \rightarrow \frac{1}{\infty} = 0$.

(d) No: $\sigma(x) \in (0, 1)$ strictly, never reaching 0 or 1.

12. Why is sigmoid useful for converting scores to probabilities?

- (a) Output always in $(0, 1)$ — valid probability range.
- (b) Input 0 gives output 0.5 — maximum uncertainty; the model has "no opinion".

13. Softmax: $\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$.

- (a) The denominator sums all e^{z_j} , ensuring all softmax outputs sum to 1 (valid probability distribution).
- (b) If $z_i \gg z_j$ for all $j \neq i$, then e^{z_i} dominates the denominator; $\text{softmax}(z_i) \rightarrow 1$.
- (c) If all z_j are equal, each e^{z_j} is equal, so each softmax value = $\frac{1}{n}$ (e.g. $\frac{1}{3}$ for three classes).

14. Learning rate decay: $\text{lr}(t) = \frac{\eta_0}{1 + \lambda t}$.

- (a) $t = 0$. $\text{lr}(0) = \frac{\eta_0}{1} = \eta_0$ — the initial rate.
- (b) $t \rightarrow \infty$. Denominator $\rightarrow \infty$; $\text{lr} \rightarrow 0$.
- (c) Start high to explore the loss landscape broadly; shrink later for precise fine-tuning. Large early steps escape local minima; small later steps converge accurately.

15. Gradient update: $w_{\text{new}} = w_{\text{old}} - \eta \cdot \frac{\partial L}{\partial w}$.

- (a) **Positive gradient.** $\frac{\partial L}{\partial w} > 0$ means increasing w increases loss. So *decrease* w .
- (b) **Large gradient.** Take a big step in parameter space.
- (c) **Small η .** Each step is tiny — conservative but stable; avoids overshooting.

16. Batch normalisation: $\hat{x} = \frac{x - \mu}{\sigma}$.

- (a) $x = \mu$. Normalised value = $\frac{0}{\sigma} = 0$.
- (b) $x = \mu + \sigma$. Normalised value = $\frac{\sigma}{\sigma} = 1$.
- (c) $\sigma \approx 0$. Division by near-zero: normalised values blow up. This is a *numerical instability* — in practice a small ε is added to the denominator.

Look ahead: The z-score (standard score) from statistics is exactly this formula. Batch normalisation applies it layer-by-layer during training to keep activations in a well-behaved range.

Part E: Exponential Growth and Decay (Problems 17–19)

Concept. Compound interest, radioactive decay, and logistic growth all reduce to fractions raised to powers. The unifying idea: a quantity that changes by a fixed *ratio* each step follows $A(t) = A_0 \cdot r^t$. If $r < 1$, decay; if $r > 1$, growth.

Method. For compound interest: substitute the given n into $(1 + \frac{1}{n})^n$. For half-life: $(\frac{1}{2})^{t/t_{1/2}}$ gives the remaining fraction. For accuracy curves: substitute t into the formula and evaluate.

Watch out. Continuous compounding gives e , not infinity. Compounding more frequently does not mean earning more and more — it converges to a limit. Similarly, radioactive material never fully disappears — it asymptotically approaches zero.

Where this leads. These are discrete approximations to e^{rt} , the continuous exponential function. Calculus connects them: $\frac{d}{dt}e^{rt} = re^{rt}$, meaning e^{rt} is the unique function that is its own derivative (scaled by r).

Explore further. Bernstein, *Against the Gods: The Remarkable Story of Risk* (1996) — the history of compound interest and probability in finance.

17. Compound interest at 100% with n compoundings per year.

(a) $n = 1$. $(1 + 1)^1 = 2$.

(b) $n = 2$. $(1 + \frac{1}{2})^2 = (\frac{3}{2})^2 = \frac{9}{4} = 2.25$.

(c) $n = 12$. $(1 + \frac{1}{12})^{12} \approx 2.613$.

(d) $n = 365$. $(1 + \frac{1}{365})^{365} \approx 2.714$.

(e) Continuous compounding: the limiting value is $e \approx 2.718$. "Every instant" the interest is recalculated, but the result converges.

18. Half-life: remaining = initial $\times (\frac{1}{2})^{t/t_{1/2}}$.

(a) After 1 half-life: $(\frac{1}{2})^1 = \frac{1}{2}$ remains.

(b) After 2: $(\frac{1}{2})^2 = \frac{1}{4}$.

(c) After 10: $(\frac{1}{2})^{10} = \frac{1}{1024} \approx 0.001$, i.e. 0.1% remains.

(d) No: the exponent $t/t_{1/2}$ is always finite, so $(\frac{1}{2})^{t/t_{1/2}} > 0$ always.

19. Accuracy curve: $\text{acc}(t) = 1 - \frac{1}{1+t}$.

(a) $t = 0$. $1 - 1 = 0$.

(b) $t = 1$. $1 - \frac{1}{2} = 0.5$.

(c) $t = 9$. $1 - \frac{1}{10} = 0.9$.

(d) $t = 99$. $1 - \frac{1}{100} = 0.99$.

(e) Approaches 1 but never reaches it: $\frac{1}{1+t} \rightarrow 0$ as $t \rightarrow \infty$.

Part F: Classification Metrics (Problems 20–23)

Concept. Precision, recall, and F1 score are all fractions built from four counts: true positives (TP), true negatives (TN), false positives (FP), false negatives (FN).

Precision: of our positive predictions, how many were right? **Recall:** of all actual positives, how many did we find? **F1:** harmonic mean of the two.

Method. Substitute the given counts directly into the formulas. For qualitative questions: think about which count changes and whether that makes the fraction larger or smaller.

Watch out. You cannot maximise both precision and recall simultaneously. Predicting everything as positive maximises recall (you catch everything) but destroys precision (everything is a "positive prediction"). F1 score forces a balance.

Where this leads. In practice, you plot a **precision-recall curve** by varying the classification threshold and observe the trade-off. The area under this curve (AUPRC) summarises model quality for imbalanced datasets better than accuracy.

Explore further. Murphy, *Machine Learning: A Probabilistic Perspective* (2012), Ch. 5 for a thorough treatment of evaluation metrics.

20.
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$

(a) **90 TP, 10 FP.** $\frac{90}{100} = 0.9$.

(b) **More FP.** Denominator grows, numerator stays constant: precision decreases.

(c) **Perfect precision.** $\text{FP} = 0 \Rightarrow \frac{\text{TP}}{\text{TP}} = 1$.

21.
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

(a) **90 TP, 10 FN.** $\frac{90}{100} = 0.9$.

(b) **Predict everything positive.** No false negatives: $\text{FN} = 0$, $\text{recall} = 1$.

(c) **Precision.** Predicting everything positive means all negatives become FP: precision plummets.

22.
$$\text{F1} = \frac{2P \cdot R}{P + R}.$$

(a) $P = R = 1$. $\frac{2}{2} = 1$. Perfect.

(b) $P = 1$, $R = 0$. $\frac{0}{1} = 0$.

(c) $P = R = 0.6$. $\frac{2 \times 0.36}{1.2} = \frac{0.72}{1.2} = 0.6$.

(d) The harmonic mean penalises imbalance heavily: F1 of (1, 0) is 0, while the arithmetic mean would be 0.5. A model cannot "hide" bad recall behind good precision.

23. Accuracy on 99% "not spam" data.

- (a) Model predicting "not spam" always: accuracy = $\frac{99}{100} = 99\%$.
- (b) Zero spam emails caught.
- (c) 99% accuracy with zero utility for the task. Accuracy is misleading when classes are imbalanced: the model is rewarded for ignoring the rare (important) class entirely.

Look ahead: This is why F1, precision, and recall are preferred over accuracy for fraud detection, medical diagnosis, and spam filtering — all cases where the positive class is rare but important.

Part G: Geometric Series and Convergence (Problems 24–25)

Concept. $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ for $|r| < 1$. The formula is a fraction whose denominator is $1 - r$. When $r \geq 1$, the denominator approaches 0 or becomes negative, and the series diverges (partial sums grow without bound).

Method. Identify the first term a and ratio r . Apply $S = \frac{a}{1-r}$ for infinite sums. For expected value: $E[\text{tries}] = \frac{1}{p}$ where p is the probability of success on each trial.

Watch out. The formula $\frac{1}{1-r}$ requires $|r| < 1$ strictly. For $r = 1$: each term is 1, so the sum is $1 + 1 + 1 + \dots = \infty$. For $r = -1$: the partial sums oscillate between 0 and 1, never converging.

Where this leads. These series are the foundations of Taylor and Maclaurin series. For instance, $\frac{1}{1-x} = 1 + x + x^2 + \dots$ is both a geometric series and the Taylor expansion of $\frac{1}{1-x}$ around $x = 0$.

Explore further. Zeno's paradox of Achilles and the tortoise is resolved by a geometric series: infinitely many steps can sum to a finite total. See Stewart, *Calculus*, Ch. 11.

24. Geometric series $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$.

(a) $r = \frac{1}{2}$. $1 + \frac{1}{2} + \frac{1}{4} + \dots = \frac{1}{1-\frac{1}{2}} = 2$.

(b) $r = \frac{1}{3}$. $\frac{1}{1-\frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2} = 1.5$.

(c) $r = \frac{2}{3}$. $\frac{1}{1-\frac{2}{3}} = \frac{1}{\frac{1}{3}} = 3$.

(d) When $r \geq 1$: terms do not shrink, so partial sums grow without bound. The formula would give $\frac{1}{1-1} = \frac{1}{0}$, which is undefined.

25. Expected tries = $\frac{1}{P(\text{success})}$.

(a) Die, $p = \frac{1}{6}$. Expected tries = 6.

(b) $p = \frac{1}{100}$. Expected tries = 100.

(c) $p \rightarrow 0$. Expected tries $\rightarrow \infty$. Rare events require many attempts on average.

Part H: Challenge Problems (Problems 26–28)

Concept. These three problems combine everything: the logistic growth model mixes exponential decay in the denominator with a carrying-capacity fraction; the normal distribution uses a fraction in an exponent; and attention is just softmax applied to similarity scores.

Method. For each: identify the fraction structure, apply the limiting behaviour rules from Parts A–G. Do not be intimidated by unfamiliar symbols — cover up everything except the fraction you are analysing.

Watch out. In the normal distribution (problem 27), the fraction $\frac{(x-\mu)^2}{2\sigma^2}$ appears in an exponent. The key: any positive number in the exponent, preceded by a minus sign, makes $e^{-(\cdot)}$ smaller. Far from the mean, the exponent is large and negative, so the density is very small.

Where this leads. Logistic growth (problem 26) is the S -curve used to model technology adoption, epidemic spread, and neural network training curves. The attention mechanism (problem 28) is the core of the Transformer architecture underlying GPT, Claude, and BERT.

Explore further. Vaswani et al., *Attention Is All You Need* (2017) — the original Transformer paper. Freely available on arXiv.

26. Logistic growth:
$$P(t) = \frac{K}{1 + \frac{K-P_0}{P_0}e^{-rt}}.$$

- (a) As $t \rightarrow \infty$. $e^{-rt} \rightarrow 0$ (since $r > 0$).
- (b) **Denominator.** $1 + \frac{K-P_0}{P_0} \cdot 0 = 1$.
- (c) $P(t)$. $P \rightarrow \frac{K}{1} = K$. The population approaches the **carrying capacity**.
- (d) $t = 0$. $P(0) = \frac{K}{1+(K-P_0)/P_0} = \frac{K}{\frac{K}{P_0}} = P_0$. ✓

27. Normal distribution exponent:
$$-\frac{(x-\mu)^2}{2\sigma^2}.$$

- (a) **At $x = \mu$.** $(x-\mu)^2 = 0$; exponent = 0; $e^0 = 1$. The density is maximum at the mean.
- (b) $(x-\mu)^2$ grows as x moves away from μ .
- (c) Exponent becomes more negative (larger magnitude, negative sign).
- (d) $e^{-(\text{large positive})}$ approaches 0. Density falls off rapidly far from the mean.

Note: This is why the normal distribution is "bell-shaped": maximum at the mean, decaying symmetrically in both directions.

28. Attention: softmax of scores $[10, 0, 0]$.

- (a) **High-score weight.** $\frac{e^{10}}{e^{10}+e^0+e^0} = \frac{e^{10}}{e^{10}+2} \approx \frac{22026}{22028} \approx 0.9999$.
- (b) **Low-score weights.** $\frac{e^0}{e^{10}+2} \approx \frac{1}{22028} \approx 0.00005$ each.
- (c) The model **attends** almost entirely to the high-scoring token and ignores the others. The fraction implements a "soft argmax": the winner gets almost all the

weight.

Look ahead: The temperature parameter T in $\text{softmax}(z_i/T)$ controls sharpness: small T makes the winner-takes-all effect stronger; large T spreads weights evenly.